



Important Formulas: Finding Common Ground

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This chapter collects the important ideas and formulas you need to find common ground between two or more numbers: their factors, multiples, Highest Common Factor (HCF) and Lowest Common Multiple (LCM). Read each definition, follow the worked examples and practise the methods shown. Key terms are highlighted for quick reference.

Understanding Multiples and Common Multiples

Multiple - A multiple of a number is obtained by multiplying that number by 1, 2, 3, 4, ...

Multiples of 6: 6, 12, 18, 24, 30 ...

Multiples of 8: 8, 16, 24, 32, 40 ...

A **common multiple** of two numbers is a number that appears in both lists of multiples. Example: 24, 48, ... are common multiples of 6 and 8.

The **smallest common multiple** is called the **LCM** (Lowest Common Multiple).

Patterns with HCF when One Number is a Factor of the Other

When one number is a factor of the other, the **Highest Common Factor (HCF)** is the smaller number. Examples:

(6, 18)
(14, 42)
(12, 36)

In each case HCF = the smaller number.

Factors and Common Factors

Factor - A number is a factor of another if it divides the second number exactly (without remainder). Example: 4 is a factor of 12 because $12 \div 4 = 3$.

Factors

$$\textcircled{2} \times \textcircled{6} = \textcircled{12}$$

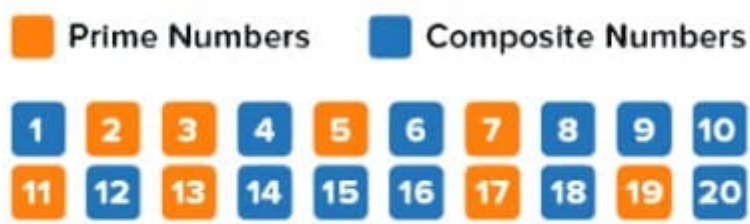
Product

Common factor - A number that divides each of the numbers in a set. Example: 4 is a common factor of 12 and 16 because $12 \div 4 = 3$ and $16 \div 4 = 4$.

Prime and Composite Numbers

Prime number - A number greater than 1 that has exactly two distinct factors: 1 and itself. Example: 7 is prime because only 1 and 7 divide it evenly.

Composite number - A number greater than 1 that has more than two factors. Example: 12 is composite because it can be divided by 1, 2, 3, 4, 6 and 12.



21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

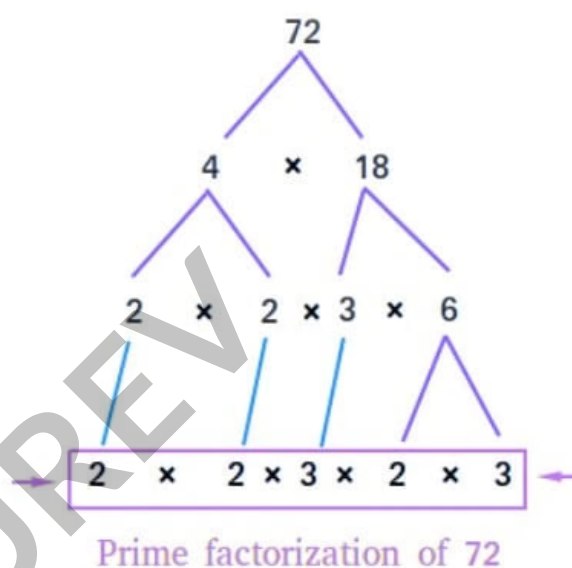
Prime Factorisation

Every integer greater than 1 can be expressed as a product of prime numbers. This expression is unique except for the order of factors. Example:

$$84 = 2 \times 2 \times 3 \times 7$$

2	72
2	36
2	18
3	9
3	3
	1

Prime Factorization of $72 = 2 \times 2 \times 2 \times 3 \times 3$
 $= 2^3 \times 3^2$



Prime factorisation is the main tool used to find HCF and LCM precisely.

Co-prime Numbers

Co-prime (or relatively prime) - Two numbers are co-prime if their only common factor is 1.

To check co-primality, find prime factorisations and see whether they share any common prime.

Example:

$$80 = 2 \times 2 \times 2 \times 2 \times 5$$

$$63 = 3 \times 3 \times 7$$

Since 80 and 63 share no common primes, they are co-prime.

Using Prime Factorisation to Find the HCF

Procedure:

Write prime factorisations of both numbers.

Find the common prime factors.

For each common prime, take its minimum power (the smaller exponent) occurring in the two factorisations.

Multiply those chosen prime powers to get the HCF.

Example: Find the HCF of 45 and 75.

Sol.

$$45 = 3 \times 3 \times 5$$

$$75 = 3 \times 5 \times 5$$

Common prime factors: 3 and 5.

Minimum power of 3 is one 3; minimum power of 5 is one 5.

$$\text{HCF} = 3 \times 5$$

$$\text{HCF} = 15$$

Using Prime Factorisation to Find the LCM

Procedure:

Write prime factorisations of both numbers.

List all primes that appear in either factorisation.

For each prime, take its maximum power (the larger exponent) occurring in the two factorisations.

Multiply those chosen prime powers to get the LCM.

Example: Find the LCM of 14 and 35.

Sol.

$$14 = 2 \times 7$$

$$35 = 5 \times 7$$

Primes required: 2 (max power 1), 5 (max power 1), 7 (max power 1).

$$\text{LCM} = 2 \times 5 \times 7$$

$$\text{LCM} = 70$$

Relationship Between HCF, LCM and Product

A very useful identity links HCF, LCM and the product of two numbers.

$$\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$$

Why this holds (brief reason):

HCF contains each common prime with the minimum exponent that appears in the factorisations.

LCM contains every prime that appears in either number with the maximum exponent. Multiplying HCF and LCM reconstructs the full prime powers necessary to produce the product of the two original numbers.

Efficient Division Method to Find HCF and LCM Together

This method divides both numbers by any common factor (not necessarily prime) repeatedly until the remaining pair are co-prime. It is often faster than factorising large numbers.

Procedure (division table method):

Arrange the two numbers side by side.

Choose any common divisor (2, 3, 5, 10, etc.) and divide both numbers by it; record the divisor in a column.

Repeat the process on the resulting pair until no common divisor greater than 1 remains.

Multiply all recorded divisors to obtain the HCF.

Multiply the HCF by the remaining (final row) numbers to obtain the LCM.

Example: Find HCF and LCM of 48 and 180 using the division method.

Sol.

Divide both numbers by 2:

$$48 \div 2 = 24$$

$$180 \div 2 = 90$$

Divide by 2 again:

$$24 \div 2 = 12$$

$$90 \div 2 = 45$$

Divide by 3:

$$12 \div 3 = 4$$

$$45 \div 3 = 15$$

Divide by 5:

$4 \div 5$ is not exact, so choose next common factor (both are not divisible by 5). Choose (both divisible by 1 only). Stop when no common divisor > 1 remains. The recorded divisors are 2, 2, 3.

Multiply recorded divisors to get HCF:

$$\text{HCF} = 2 \times 2 \times 3$$

$$\text{HCF} = 12$$

Multiply HCF with the final row numbers to find LCM:

Final row numbers are 4 and 15 (the results remaining after division by common factors).

$$\text{LCM} = \text{HCF} \times 4 \times 15$$

$$\text{LCM} = 12 \times 4 \times 15$$

$$\text{LCM} = 720$$

Euclid's Algorithm (Optional Fast Method for HCF)

Euclid's algorithm finds HCF by repeated division of the larger number by the smaller and replacing until remainder becomes zero. The last non-zero remainder is the HCF.

Example: HCF of 84 and 30 using Euclid's algorithm.

Sol. Divide 84 by 30; remainder is 24.

Replace 84 by 30 and 30 by 24; divide 30 by 24; remainder is 6.

Replace 30 by 24 and 24 by 6; divide 24 by 6; remainder is 0.

The last non-zero remainder is 6, so $\text{HCF} = 6$.

Worked Example Combining HCF and LCM

Example: For the numbers 18 and 24, find HCF and LCM.

Sol.

Prime factorise each number.

$$18 = 2 \times 3 \times 3$$

$$24 = 2 \times 2 \times 2 \times 3$$

Common primes: 2 and 3. Minimum powers give HCF; maximum powers give LCM.

$$\text{HCF} = 2 \times 3 = 6$$

$$\text{LCM} = 2^3 \times 3^2 = 8 \times 9$$

$$\text{LCM} = 72$$

Generalisation and Conjectures

Conjecture: A claim made after observing patterns; it needs proof to become a theorem.

Generalisation: A statement or rule that holds for all cases of a given type after proof; for example the identity linking HCF and LCM with product is a general result for any two positive integers.

Practice Problems

List the first five multiples of 9 and 12. Find their LCM.

Find the HCF of 56 and 98 by prime factorisation.

Use Euclid's algorithm to find the HCF of 101 and 462.

Check whether 35 and 48 are co-prime.

Final Summary

Prime factorisation is the most reliable method to find HCF and LCM. For HCF take common primes with minimum powers; for LCM take all primes with maximum powers. The identity

$$\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$$

helps to check answers or compute one value if the other and the product are known. The division table method and Euclid's algorithm are efficient alternatives for larger numbers. Practise these methods to become quick and accurate at finding common factors and multiples.

